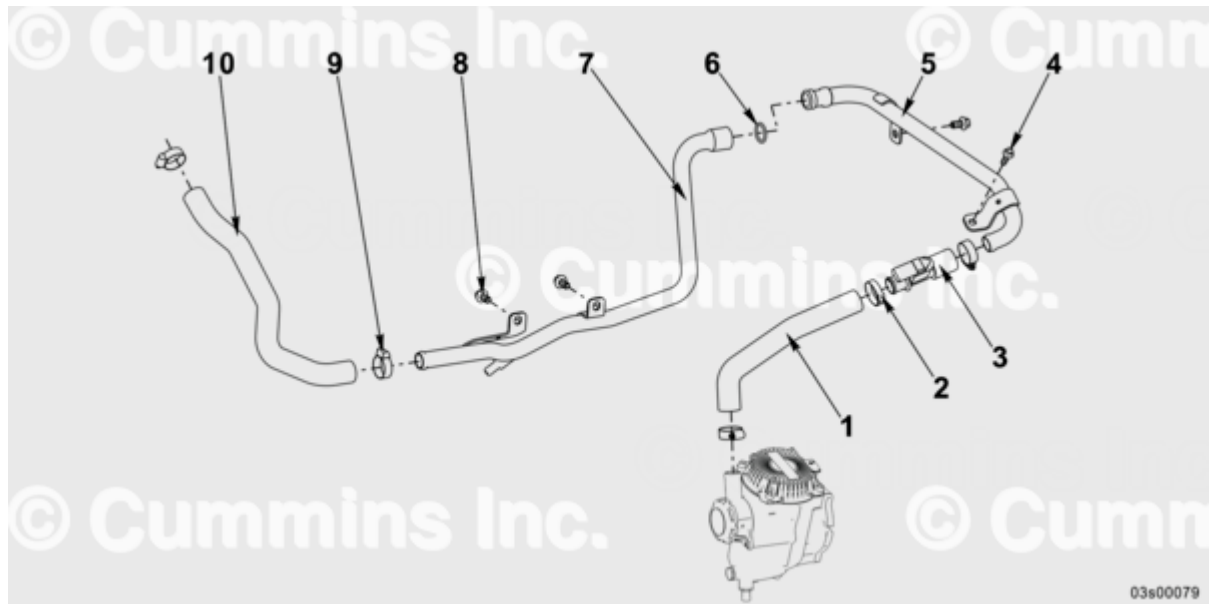


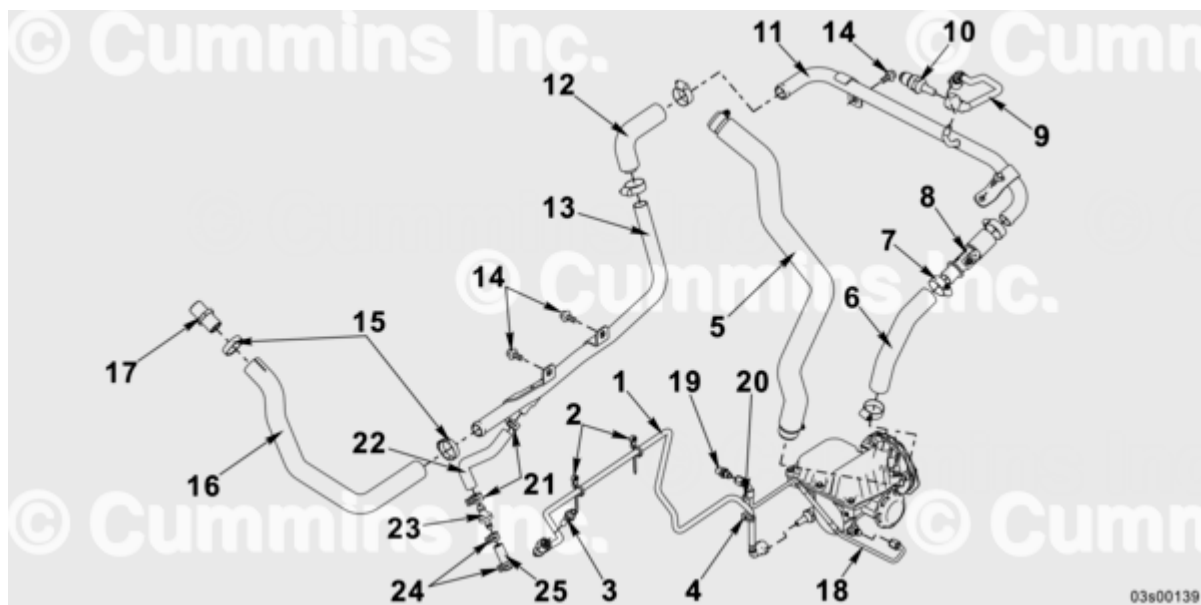
Trainings ▾

Exploded View



Electric Drive Closed Crankcase Ventilation (CCV) Hoses Exploded View

1. Hose - quantity 1
2. Clamp - quantity 1
3. Crankcase breather adapter - quantity 1
4. Capscrew M8 - quantity 1
5. Crankcase breather tube - quantity 1
6. O-ring - quantity 1
7. Crankcase breather tube - quantity 1
8. Capscrew M10 - quantity 3
9. Clamp - quantity 4
10. Hose - quantity 1



Air Drive CCV Hoses Exploded View

1. Boost air tube - quantity 1
2. Clamp - quantity 2
3. Quick-disconnect connector - quantity 1
4. Clamp - quantity 1
5. Crankcase breather tube - quantity 1
6. Crankcase breather tube - quantity 1
7. Clamp - quantity 7
8. Crankcase breather adapter - quantity 1
9. Bypass tube - quantity 1
10. Check valve - quantity 1
11. Crankcase breather tube - quantity 1
12. Crankcase breather tube - quantity 1
13. Crankcase breather tube - quantity 1
14. Capscrew - quantity 3
15. Clamp - quantity 2
16. Crankcase breather tube - quantity 1
17. Hose adapter - quantity 1
18. Oil supply line - quantity 1
19. Male union - quantity 1
20. Clip - quantity 1
21. Clamp - quantity 2
22. Oil drain tube - quantity 1
23. Check valve - quantity 1
24. Clamp - quantity 2
25. Oil drain tube - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

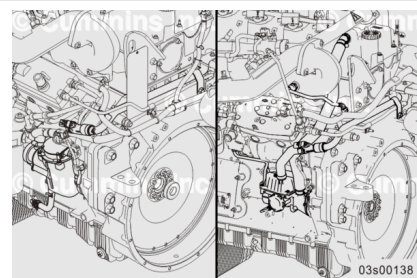
- No recommended Cummins® service tools required

Additional Service Items

- No additional service items required.

General Information

The crankcase breather system is a closed system. Oil vapor is separated by the crankcase breather motor or the crankcase breather filter, condensed in the breather housing, and is returned to the oil pan by external plumbing or the flange face in the breather housing. The crankcase gases vent to the turbocharger compressor inlet.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

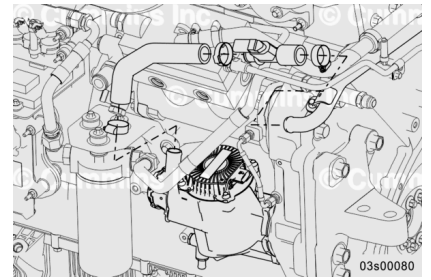
- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the CCV system pressure sensor.
- Disconnect the oil drain hose from the air transfer tube. Refer to Procedure 003-001 in Section 3. (</qs3/pubsys2/xml/en/procedures/1016/1016-003-001.html>)

Remove

For the electric drive crankcase breather:

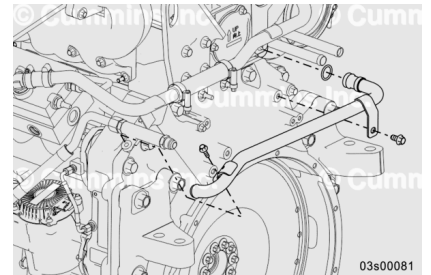
Remove the CCV hose and adapter from the CCV motor outlet and the CCV tube at the rear of the engine.

Remove the ventilation hose from the adapter and discard the clamp between the ventilation hose and adapter.



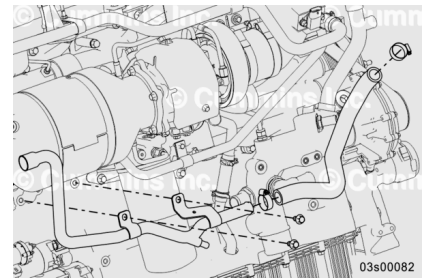
Remove the ventilation tube from the rear of the engine and the right-side ventilation tube.

Remove and discard the o-ring.



Remove the crankcase ventilation hose from the turbocharger compressor inlet and right-side crankcase ventilation tube.

Remove the right-side crankcase ventilation tube from the engine.

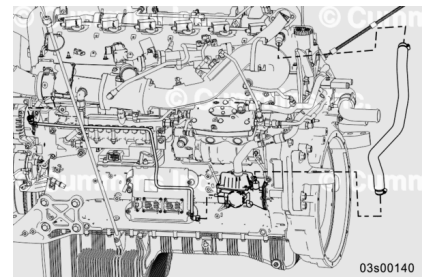


For the air drive crankcase breather:

Remove the boost air tube from the quick-disconnect connector in air connection adapter and the crankcase breather. Remove the quick-disconnect connector if needed.

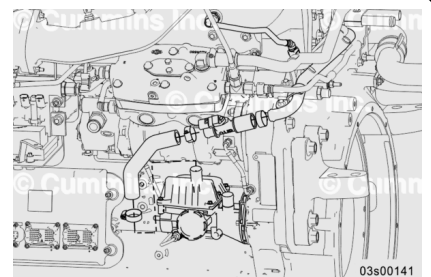
Remove the oil supply line from cylinder block and crankcase breather.

Remove the crankcase breather tube from rocker lever cover and the crankcase breather inlet at the rear of the engine.



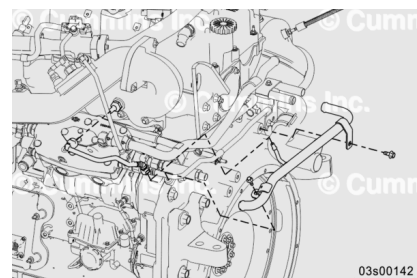
Remove the crankcase breather tube from the crankcase breather outlet.

Remove the adapter and the crankcase pressure sensor. Discard the clamps.

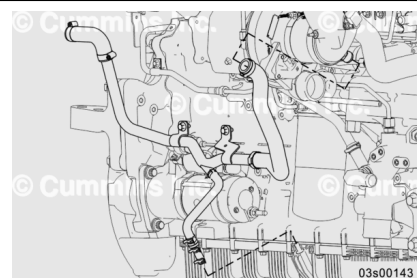


Remove the bypass tube. Remove the check valve if necessary.

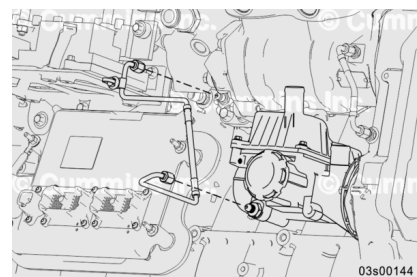
Remove the crankcase breather tube from the rear of the engine.



Remove the crankcase breather tubes from the turbocharger compressor inlet and oil pan.



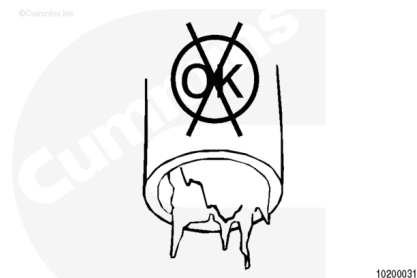
Remove the oil supply line from cylinder block and crankcase breather.



Clean and Inspect for Reuse

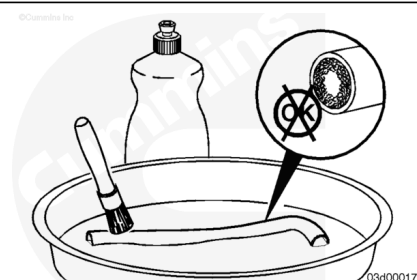
Inspect the breather hoses for sludge, debris, or ice in the tube.

Inspect the tube more frequently in icy conditions.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ CAUTION ⚠

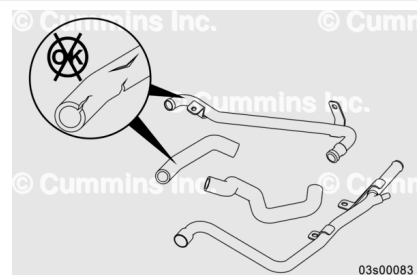
Never apply more than 34.5 kPa [5 psi] to any tube with a check valve. Unregulated shop air pressure can damage the check valve internally.

If sludge, debris, or ice is found, clean the tube with detergent and warm water or solvent.

Dry the tube with compressed air. Never apply more than 34.5 kPa [5 psi] to any tube with a check valve.

Inspect hoses and tubes for cracks or other damage.

If damage is found, replace the closed crankcase breather tube.



Install

For the electric drive crankcase breather:

Install the CCV tube onto the right side of the engine.

Loosely install the tube fasteners.

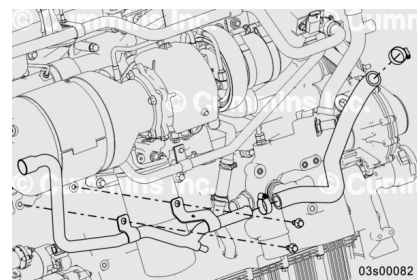
Install the CCV hose between the tube and the turbocharger compressor inlet.

Tighten the hose clamps.

Torque Value: 5 n•m [44 in-lb]

Tighten the ventilation tube fasteners.

Torque Value: 47 n•m [35 ft-lb]

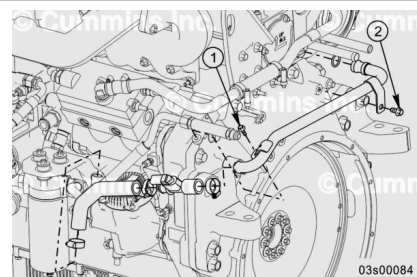


Install the crankcase ventilation tube onto the rear of the engine and onto the right-side ventilation tube. Use a new o-ring.

Loosely install the mounting nuts.

Install the ventilation hose and adapter onto the ventilation motor outlet and the rear ventilation tube.

Use a new clamp between the ventilation hose and adapter.



Tighten the hose clamps.

Torque Value: 5 n•m [44 in-lb]

Tighten the tube mounting nuts.

Torque Value:

M8 (1) 24 n•m [212 in-lb]

Torque Value:

M10 (2) 47 n•m [35 ft-lb]

For the air drive crankcase breather:

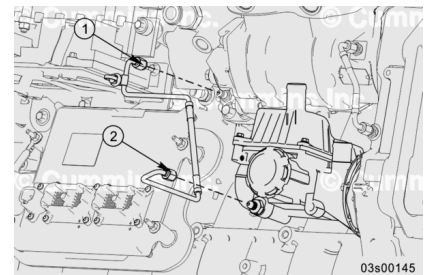
Install the oil supply line onto cylinder block and crankcase breather.

Torque Value:

(1) 25 n•m [221 in-lb]

Torque Value:

(2) 20 n•m [177 in-lb]



Install the crankcase breather tubes onto the exhaust side of the engine.

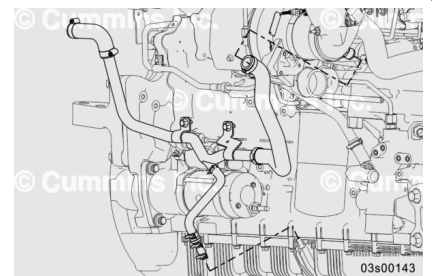
Loosely install the tube fasteners.

Tighten the hose clamps.

Torque Value: 5 n•m [44 in-lb]

Tighten the tube fasteners.

Torque Value: 47 n•m [35 ft-lb]

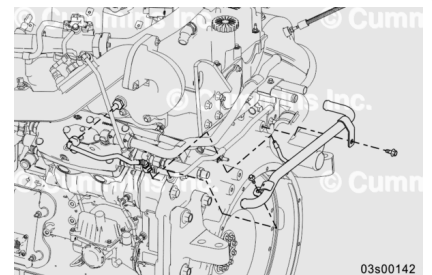


Install the crankcase breather tube onto the rear of the engine.

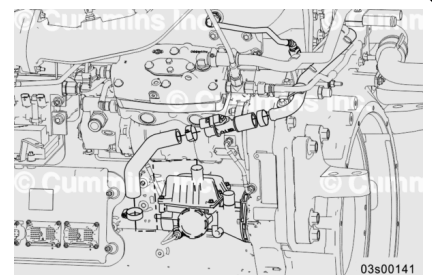
Loosely install the mounting nuts.

Install the bypass tube. Install the check valve if removed.

Torque Value: 55 n•m [41 ft-lb]

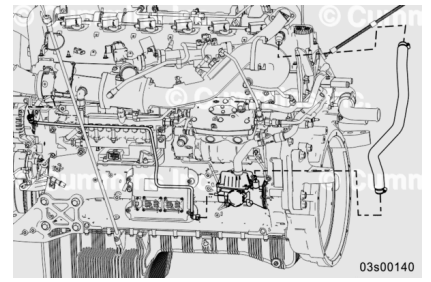


Install the crankcase breather tube and adapter onto the crankcase breather outlet.



Install the crankcase breather tube onto the rocker lever cover and the crankcase breather inlet.

Install the boost air tube. Install the quick-disconnect connector into air connection adapter if removed.



Torque Value: 16 n•m [142 in-lb]

Tighten the hose clamps.

Torque Value: 5 n•m [44 in-lb]

Tighten the tube mounting nuts.

Torque Value:

M8 (1) 24 n•m [212 in-lb]

Torque Value:

M10 (2) 47 n•m [35 ft-lb]

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the oil drain hose to the air transfer tube. Refer to Procedure 003-001 in Section 3. (</qs3/pubsys2/xml/en/procedures/1016/1016-003-001.html>)
- Connect the CCV system pressure sensor.
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.